X-ray analysis on pigments

First measurements

Linda Biasci

08/06/2026

Introduction

X-ray transmission is measured on inorganic pigments: acquisitions of counts are performed in order to verify the different behaviour of pigments based on different elements and to estimate each pigment's thickness.

Materials

All analysed pigments are mixed with an organic binder or an inorganic matrix (calcium carbonate or calcium sulphate), which have a negligible influence on X-ray analysis, and spread on a wood piece of width $5.0 \pm 1.0\ mm$:

- mercury-based: Cinnabar (HgS)
- lead-based: Lead Carbonate ($PbCO_3$) and Naples Yellow ($Pb_2Sb_2O_7$)
- cadmium-based: Cadmium Yellow (CdS)
- organic substances: egg on wood, madder lake on wood

Experimental setup

The X-ray detector is placed next to the sample (behind the sample for transmission measurements, at an approximate distance of 0.5 mm) and the distance between the source and the detector is approximately 75 cm.

Source

The employed source is a micro-focus X-ray tube (Hamamatsu), characterised by:

- tungsten anode in transmission mode
- energy range: $kVp = 120\ keV$
- anode current: $I = 20\ \mu A$
- filtration: an aluminum filter of width 1 mm, four copper filters of width 250 μm each
- acquisition time: exposition of 100 s
- collimator: a tungsten disk of thickness 2 mm, with a hole diameter of 1 mm

Detector

The employed detector is a XR-100CdTe X-Ray and Gamma Ray Detector (Amptek), characterised by:

- thickness: 1 mm
- efficiency $> 90\%$ in the range $[5\ keV, 70\ keV]$
- energy resolution: $FWHM < 1.5\ keV$
- detector area: 5x5 mm^2

Calibration

A preliminary instrumental calibration is required before starting transmission acquisitions. At first order approximation, keV to channels conversion is modeled as a linear function: the lead fluorescence spectrum is acquired and the calibration parameter is estimated by comparing the position of measured peaks to the known energy position of peaks (approximately, a conversion parameter $[ch]/[keV] \sim 7.5$ is expected).
The actual calibration is performed with a linear fit on the incident spectrum, taking tungsten fluorescence peaks as data points: the used model function is $y = c \cdot x + a$, being c the conversion parameter (from keV to channels) and a a possible calibration offset. Employed data points are as follows:

- tungsten's K-α expected peaks: 57.981 keV, 59.318 keV
- tungsten's K-β expected peaks: 67.244 keV, 69.089 keV
- channels corresponding to peaks extracted from the measured incident spectrum: $y = [457, 465, 526, 542]$
- considered uncertainties: $\sigma_K = 0.001\ keV$ on known fluorescence values, $\sigma_m = 2\ keV$ on measured peak values
- fit results: calibration factor $c = 7.673 \pm 0.151$, offset $a = 10.996 \pm 9.604$, chi square / dof = 1.063/2

Methods

Acquired data is saved and converted in a `csv` file. Radiation emitted from the source is detected and the signal is plotted:

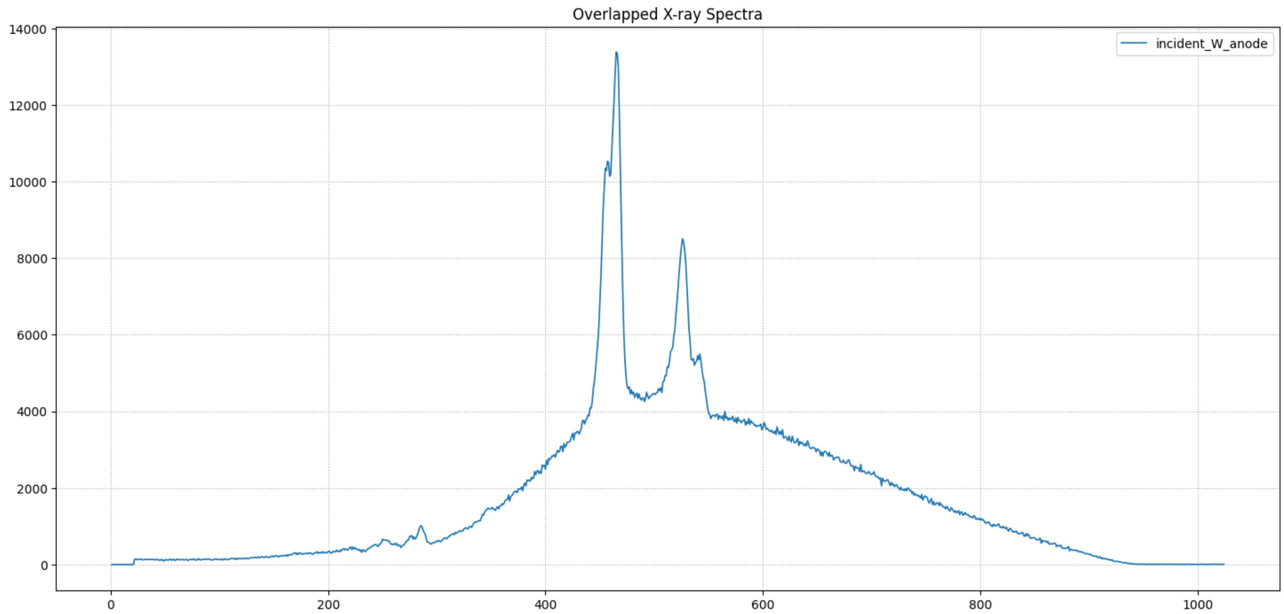

Figure 1: Measured spectrum of incident radiation: energy channels (from 1 to 1024) are represented on horizontal axis, photon counts are represented on vertical axis; the detector resolution is high enough to allow to recognise four different tungsten fluorescence peaks

Analogously, radiation transmitted through the samples is detected and respective signals are visualised, showing slightly different spectra but different degree of attenuation:

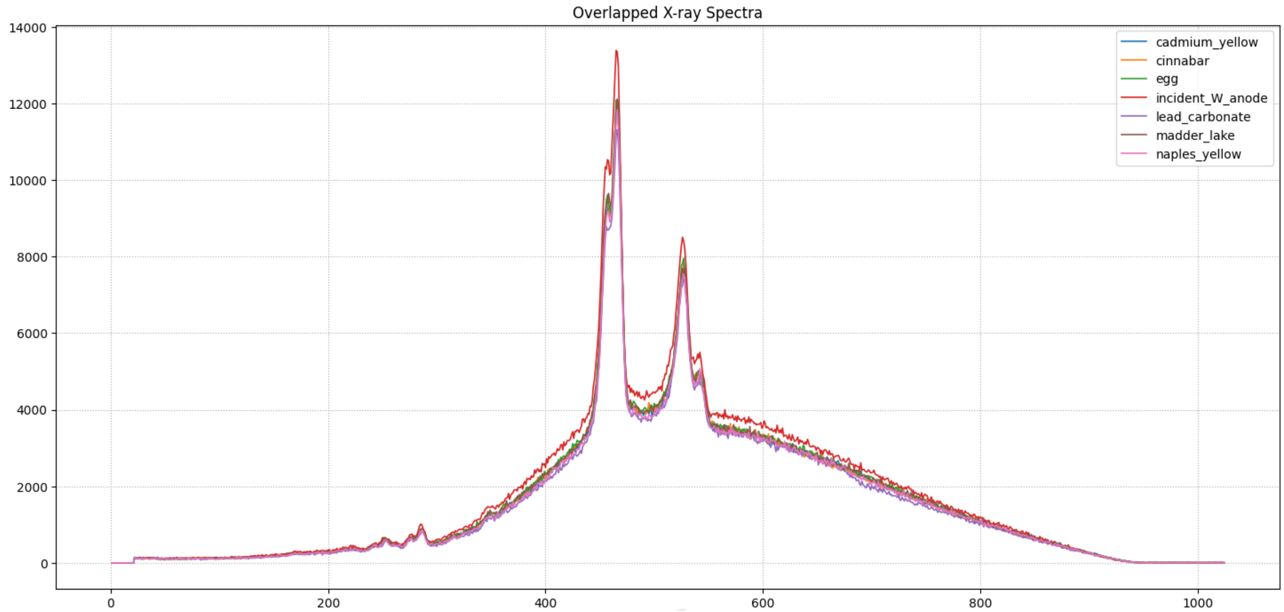

Figure 2: Measured spectrum of trasmitted radiation: analysed samples are egg and madder lake, lead carbonate and naples yellow, cinnabar, cadmium yellow; energy channels (from 1 to 1024) are represented on horizontal axis, photon counts are represented on vertical axis

Data are then analysed to produce total attenuation spectra and possibly spot K-edges of materials present in pigments and consequently derive pigments thickness. Hence, the function $-ln(I/I_0) = \sum \mu(E) \cdot x$ is calculated and plotted, being I the signal of radiation transmitted by the sample and I_0 the background or flat field signal, that could either be the incident spectrum or the signal that most resembles a pure wood spectrum. In this case, the thinnest drafts of egg and madder lake, being almost transparent to X-rays (since being thin layers of organic substances), are used to approximate the transmission spectrum of wood. For each considered pigment, three smoothed total attenuation spectra are computed, using three different I_0 signals: the incident signal, egg and madder lake. Egg and madder lake attenuation signals (computed using the flat field incident radiation as I_0) are also plotted and compared.

Thickness estimation

Total attenuation spectra of different pigments are plotted and saved in `csv` files: in the cases of lead and mercury-based pigments, the respective K-edge is looked for in the plot and the correspondent attenuation value is extracted from the saved data; in the cases of egg, madder lake and cadmium yellow (this latter not showing any measurable K-edge because of filtration), attenuation values are extracted in correspondence of the first tungsten K-α peak. In all cases, tabulated $\mu(E)$ are used to derive the pigment or wood thickness. To estimate pigments thickness, the egg over the incident radiation is taken as the I_0 signal, whereas to estimate wood thickness, contributions of egg and madder lake in attenuation are assumed negligible because of their thinness (i.e. their presence is neglected in attenuation, using $\mu(E) = \mu_{wood}(E)$).

Smoothing

A moving average smoothing via the `Python` function `numpy.convolve` with a pre-defined filter is applied to data, in order to reduce stochastic noise and improve the visualisation of experimental data.

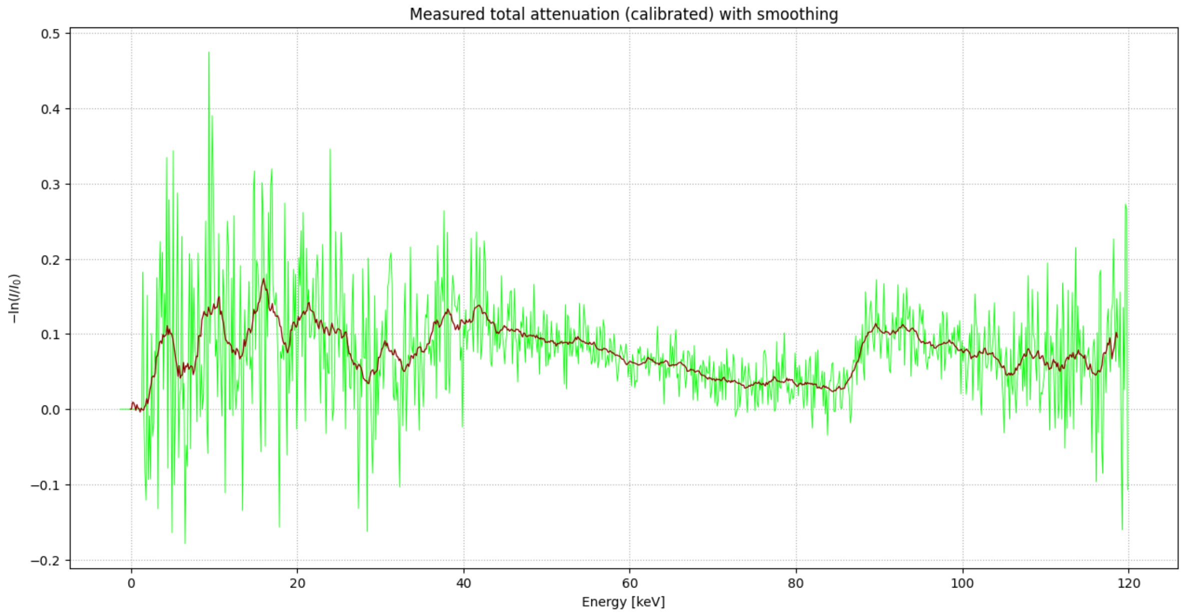

Figure 3: Example of smoothing application: total attenuation signal of lead carbonate (I) over egg on wood (I_0); the green plot shows the raw plot of raw experimental data (energy-calibrated), the red plot displays the smoothed data

Results

Organic substances

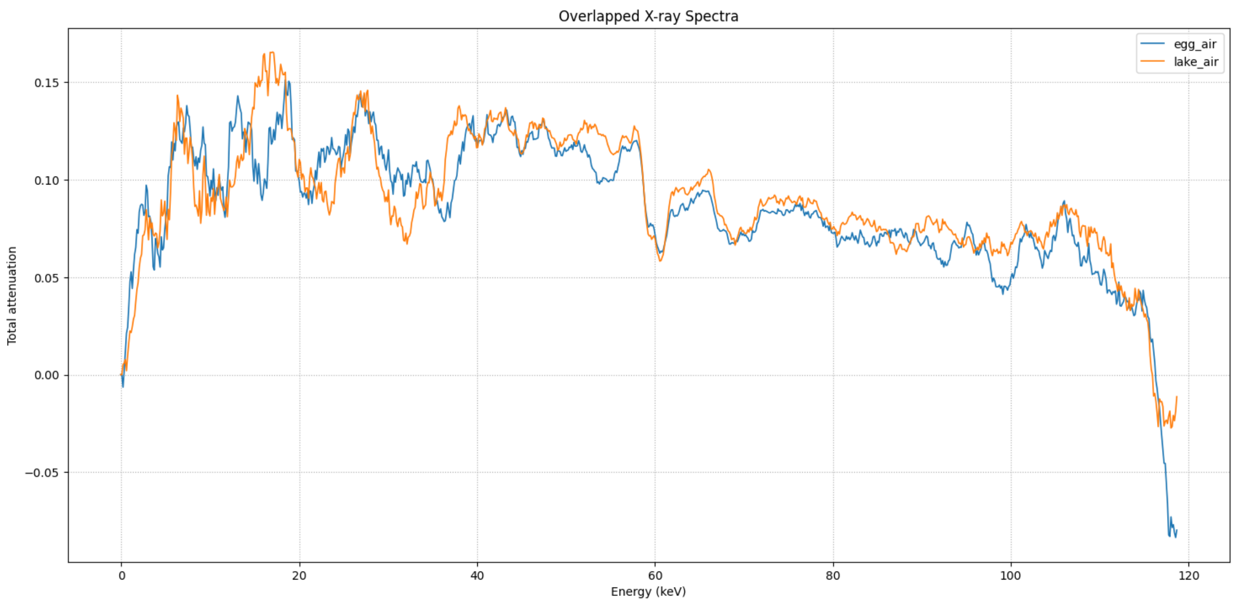

Figure 4: Comparison between egg and madder lake attenuation, $-ln(I/I_0)$, with I_0 incident radiation spectrum: the two organic substances display a similar behaviour, assumingly attributable to wood

Thickness estimation: sample's thickness (in this case, mainly wood) is obtained dividing the extracted attenuation (evaluated at the measured peak position) by the expected linear attenuation coefficient. Derived wood thickness appears to be compatible with the measured one.

Sample	Expected peak $[keV]$	Expected $\mu(E_{peak})$	Peak position $[keV]$	Attenuation $-ln(I/I_0)$	Thickness $[mm]$
Egg on wood	59.318 (tungsten)	$\sim 0.140\ cm^{-1}$	60.016	$7.151 \cdot 10^{-2}$	5.107
Madder lake on wood	59.318 (tungsten)	$\sim 0.140\ cm^{-1}$	60.016	$6.952 \cdot 10^{-2}$	4.966

Cinnabar

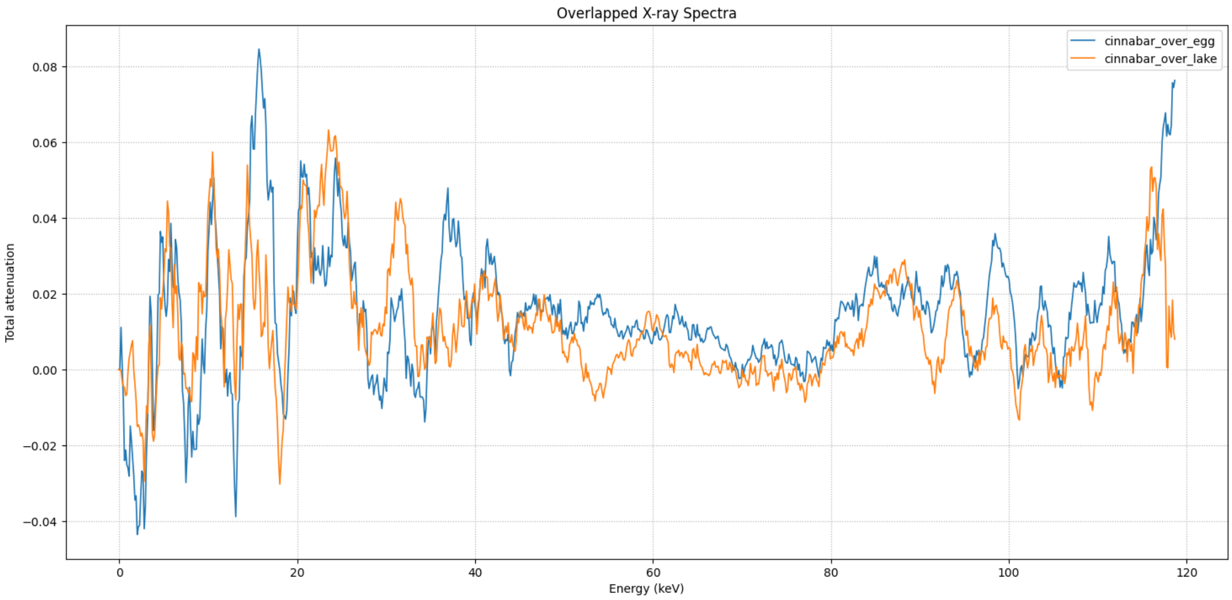

Figure 5: $-ln(I/I_0)$ signals for Cinnabar, comparing the egg signal and the madder lake signal as I_0

Despite the smoothing, the mercury's K-edge does not seem evident, an effect which can be attributed to the pigment thinness or dilution (all pigments are mixed up with organic binders or convering calcium carbonate or sulphate, but the presence of these latter cannot be recognised with K-edge analysis: they can be assumed to add a small contribution to attenuation and to be responsible for pigment dilution).
Thickness estimation (Cinnabar over egg):

Sample	Expected K-edge $[keV]$	Expected $\mu(E_K)$	Steep position $[keV]$	Attenuation $-ln(I/I_0)$	Thickness $[\mu m]$
Cinnabar	83.102 (mercury)	$114.687\ cm^{-1}$	83.909	$2.995 \cdot 10^{-2}$	2.611

Lead Carbonate

Figure 6: $-ln(I/I_0)$ signals for Lead Carbonate, comparing the egg signal and the madder lake signal as I_0

Despite the dilution, using a proper smoothing allows to clearly spot the lead's K-edge (in both cases). Thickness estimation (Lead Carbonate over egg):

Sample	Expected K-edge $[keV]$	Expected $\mu(E_K)$	Steep position $[keV]$	Attenuation $-ln(I/I_0)$	Thickness $[\mu m]$
Lead Carbonate	88.005 (lead)	87.202 cm^{-1}	88.036	$8.804 \cdot 10^{-2}$	10.096

Naples Yellow

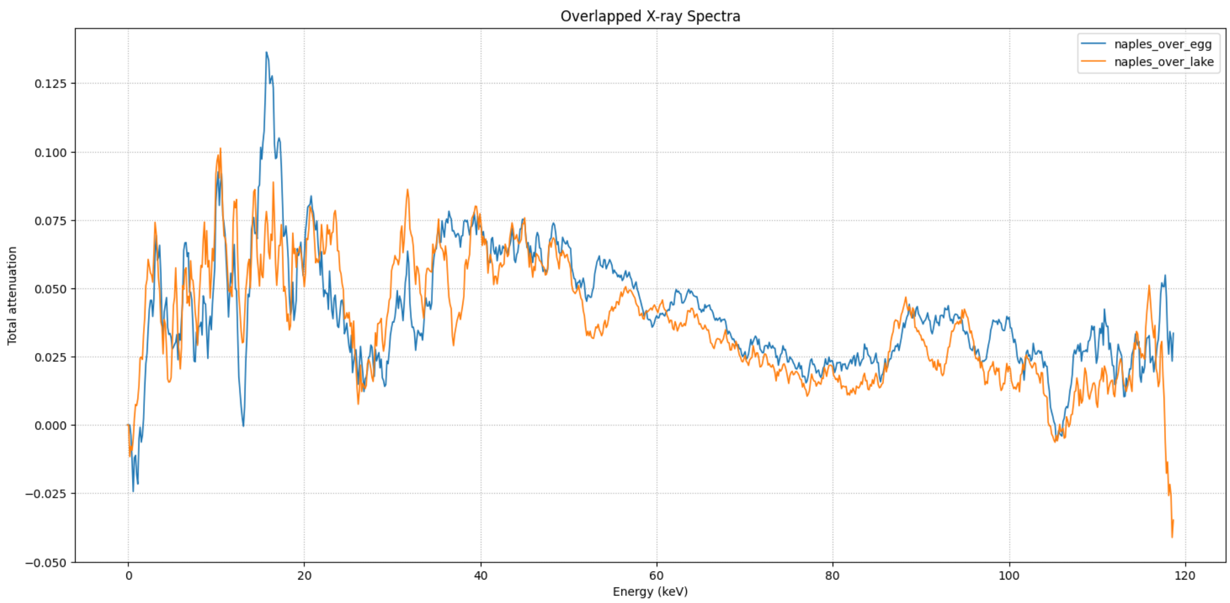

Figure 7: $-ln(I/I_0)$ signals for Naples Yellow, comparing the egg signal and the madder lake signal as I_0

Using a proper smoothing allows to identify a possible lead's K-edge trace in the pigment, despite it not being particularly evident. This might be due to a higher pigment dilution or a thinner pigment with respect to the previous (lead-based) one.
Thickness estimation (Naples Yellow over egg):

Sample	Expected K-edge $[keV]$	Expected $\mu(E_K)$	Peak position $[keV]$	Attenuation $-ln(I/I_0)$	Thickness $[\mu m]$
Naples Yellow	88.005 (lead)	87.202 cm^{-1}	88.688	$4.419 \cdot 10^{-2}$	5.068

Cadmium Yellow

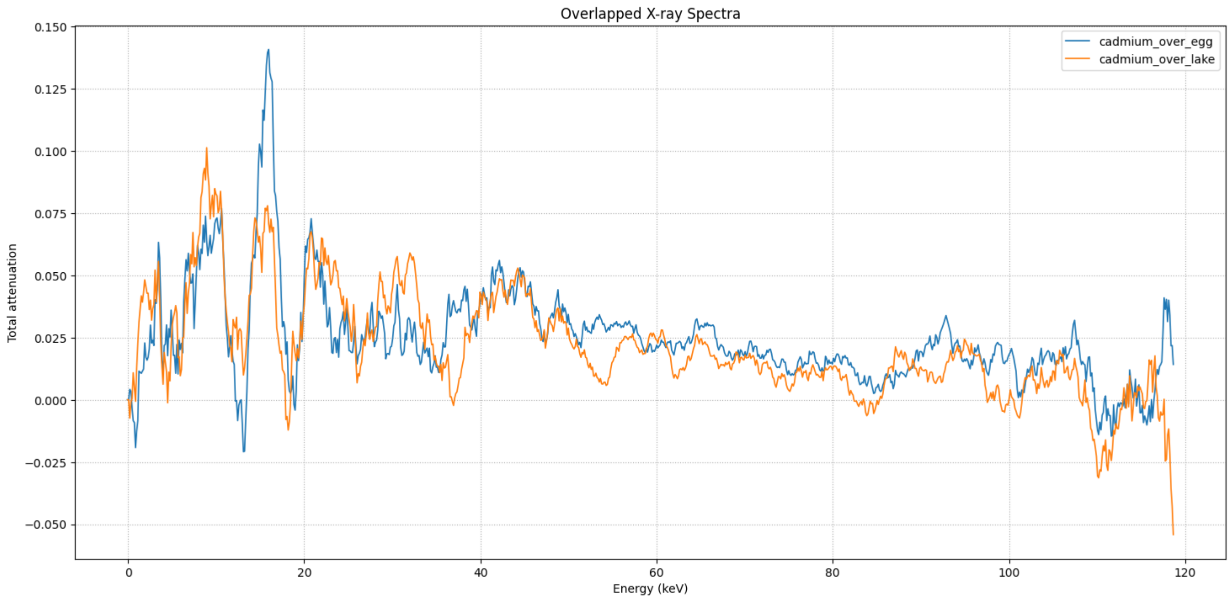

Figure 8: $-ln(I/I_0)$ signals for Cadmium Yellow, comparing the egg signal and the madder lake signal as I_0

The cadmium's K-edge (expected at 26.711 keV) cannot be identified because of copper and aluminum X-ray filtration applied to the source.
Thickness estimation (Cadmium Yellow over egg):

Sample	Expected peak $[keV]$	Expected $\mu(E_{peak})$	Peak position $[keV]$	Attenuation $-ln(I/I_0)$	Thickness $[\mu m]$
Cadmium Yellow	59.318 (tungsten)	51.684 cm^{-1}	58.886	$2.046 \cdot 10^{-2}$	3.959

Conclusions

Results obtained allow to run a correspondent simulation, using the known materials and their derived thicknesses. With such an experimental resolution, exposition time and total photon flux, pigments can be concluded to be visually recognisable if drafted in thick enough layers, approximately thicker than $\sim 100\ \mu m$. Nevertheless, lead and mercury-based pigments might be identified even at smaller widths (in the case of a thickness of at least $\sim 10\ \mu m$) employing K-edge analysis, provided that the experimental resolution is high enough not to convolve their peaks (i.e. it should be, approximately, $FWHM < 6\ keV$).

Links

[1] Amptek detectors documentation and specifications here

[2] Tables of linear attenuation coefficients and material densities from NIST here

[3] Material K-edges can be found in the X-ray data booklet here